

BEYOND SPEED RANKINGS: A NINE-COUNTRY MEASUREMENT STUDY OF SOUTHEAST ASIA’S APPLICATION-LAYER DIGITAL DIVIDE

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990M

Speed tests analyzed

9

SE Asian countries

100%

HD-video success, every country

<38%

Cloud-gaming success, weakest markets

Motivation

Speed-index rankings say little about whether real applications work. Thailand ranked **12th worldwide** for fixed broadband (Ookla Speedtest Global Index, June 2026), yet a 2023 Thailand Consumer Council survey found **81%** of 2,924 respondents had connection problems — mostly slow and dropping connections. The mismatch is not unique to Thailand.

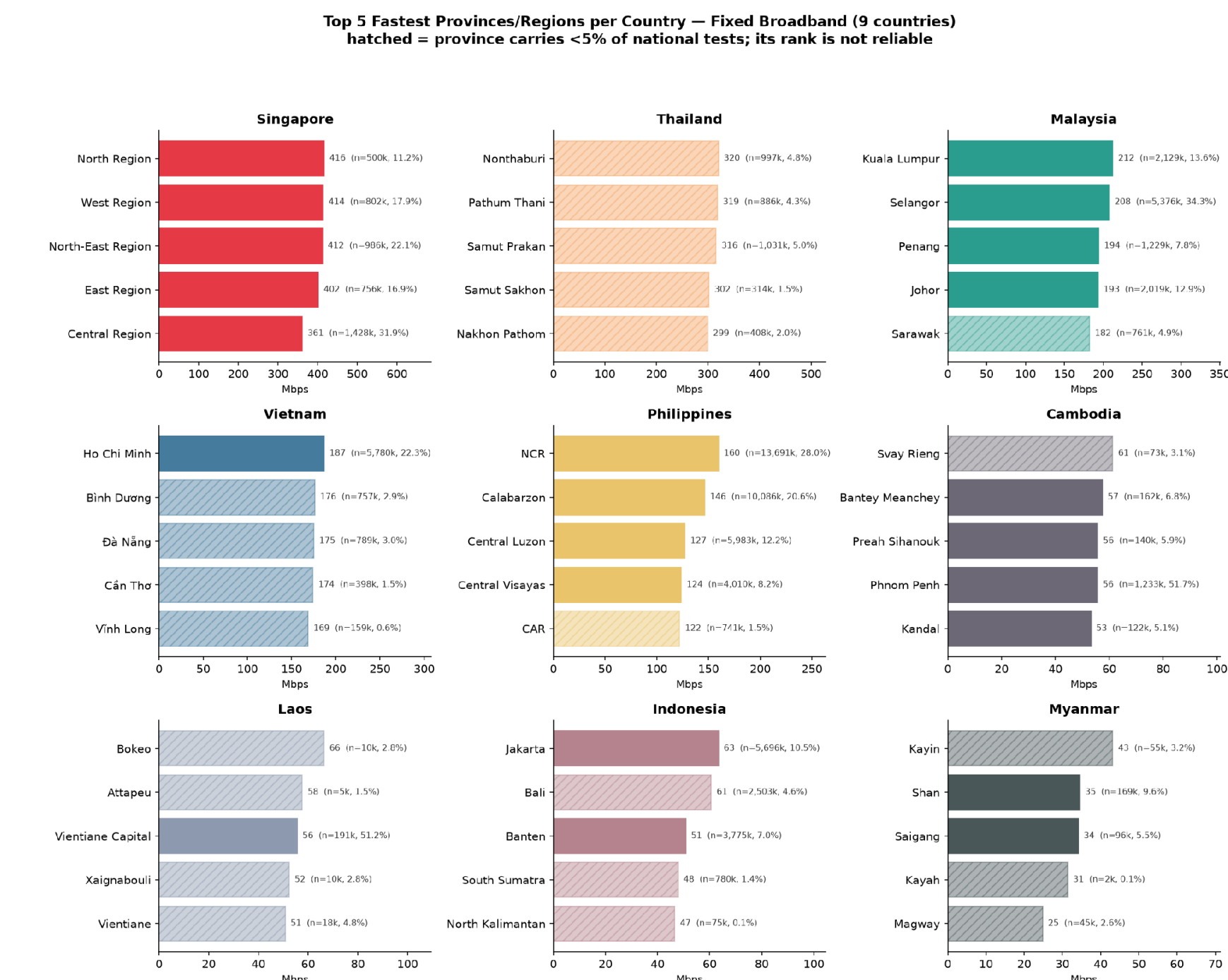
Question: how do nine Southeast Asian networks perform against the demands of real applications — voice, HD video, UHD video and cloud gaming?

Data

- **Ookla Open Data:** 229.4M tests, quarterly ~600 m tiles, fixed + mobile
- **M-Lab NDT7:** 760.6M single-stream tests
- **Coverage:** Thailand, Vietnam, Laos, Myanmar, Cambodia, Indonesia, Philippines, Malaysia, Singapore; Q1 2023–Q4 2025
- Tiles mapped to provinces (point-in-polygon, geoBoundaries ADM1), test-weighted; province-quarters with ≥ 100 tests and ≥ 5 tiles

Where the Speed Is

The five fastest provinces per country, hatched where a province carries under **5%** of its country’s tests and its rank is therefore not reliable. Singapore clears **360 Mbps** in every region; Thailand’s top five are all Bangkok-adjacent and all thin-sample. Malaysia and the Philippines concentrate in hubs that do carry real volume (Selangor 34.3%, NCR 28.0%), while the emerging markets top out below 200 Mbps and fill their rankings with low-sample provinces.



Top five provinces per country, fixed broadband; hatched = under 5% of national tests.

Data & Code



github.com/chissanapun/
thailand-broadband-analysis

Finding 1 — Basics Are Solved

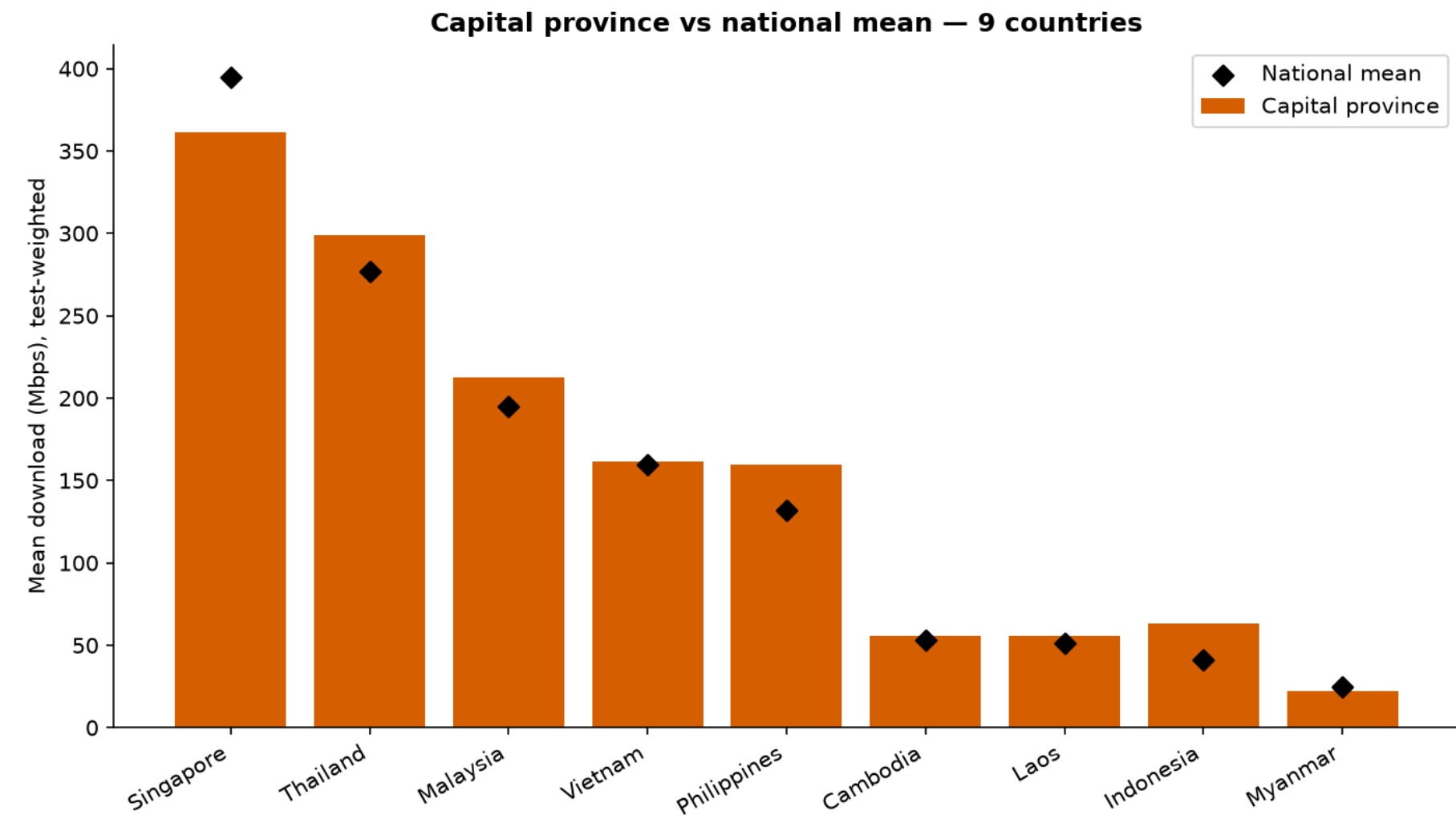
Voice (≥ 64 kbps) and HD video (≥ 5 Mbps) succeed at **100%** in every country, every quarter, on fixed and mobile networks. Even the lowest-tier broadband markets meet the basic floor.

Application	Data rate	Latency
Voice	64 kbps	200 ms
Video (HD)	5 Mbps	few s
Video (UHD)	25 Mbps	few s
Cloud gaming	44 Mbps	25 ms

Thresholds after Lübben & Misfeld (2022). Success rate = national-level share meeting the download-speed threshold, per quarter.

Finding 2 — Urban–Rural Stratification

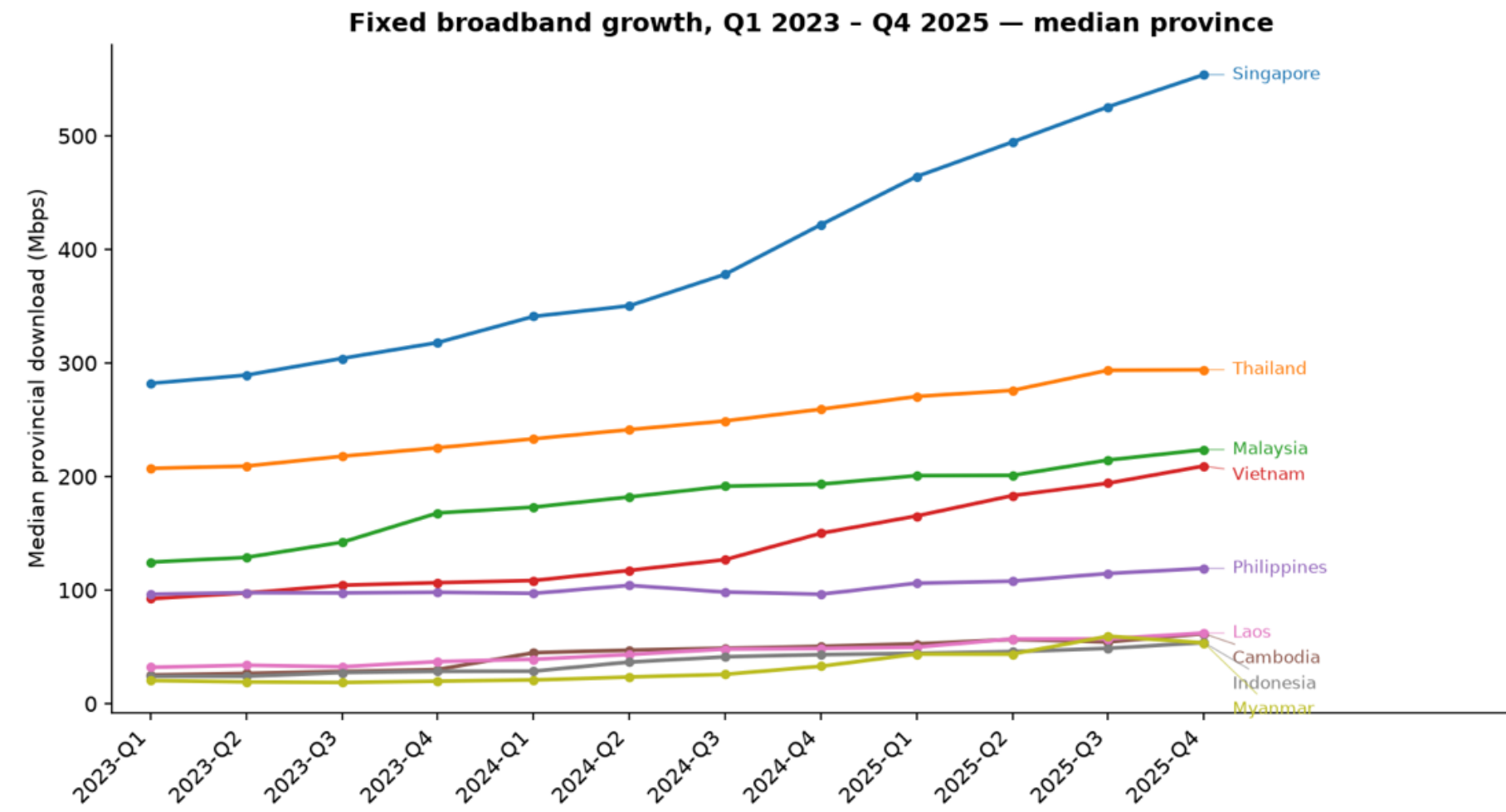
Most capitals outperform their nation: Bangkok 298.8 vs. 276.9 Mbps, Kuala Lumpur 212.3 vs. 194.8, Manila NCR 159.6 vs. 132.0. Singapore’s Central Region (361.0) sits *below* its national mean (394.8). Peak capital speed is **over 16 $\times$** the lowest capital (Myanmar, 22.0 Mbps).



Capital province vs. national mean, fixed broadband, test-weighted.

Finding 3 — Speed Gap Is Widening

Singapore nearly **doubled** its median provincial speed ($\sim 280 \rightarrow \sim 550$ Mbps); Thailand grew $\sim 200 \rightarrow \sim 300$. Vietnam accelerated to catch Malaysia (200+ Mbps). Laos, Cambodia, Indonesia and Myanmar stayed **below 60 Mbps** for all three years.

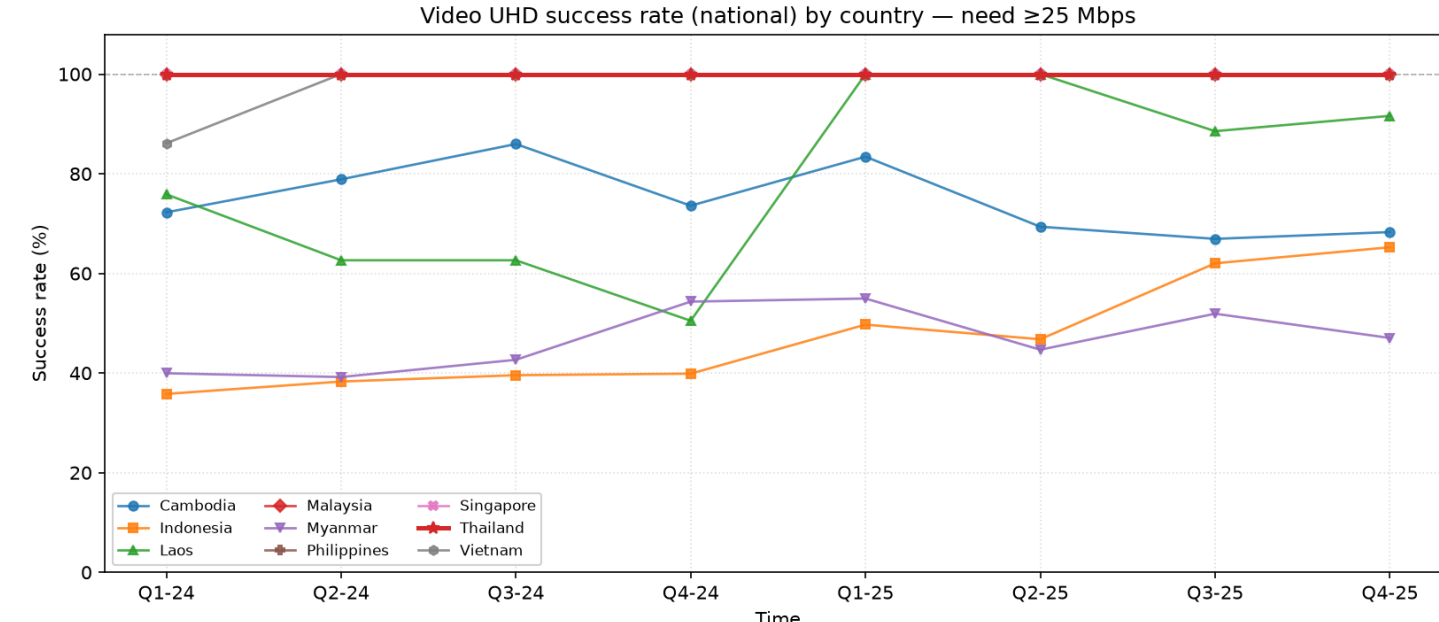


Median provincial fixed download speed, Q1 2023–Q4 2025.

Finding 4 — Application Tiers Diverge

UHD video (≥ 25 Mbps)

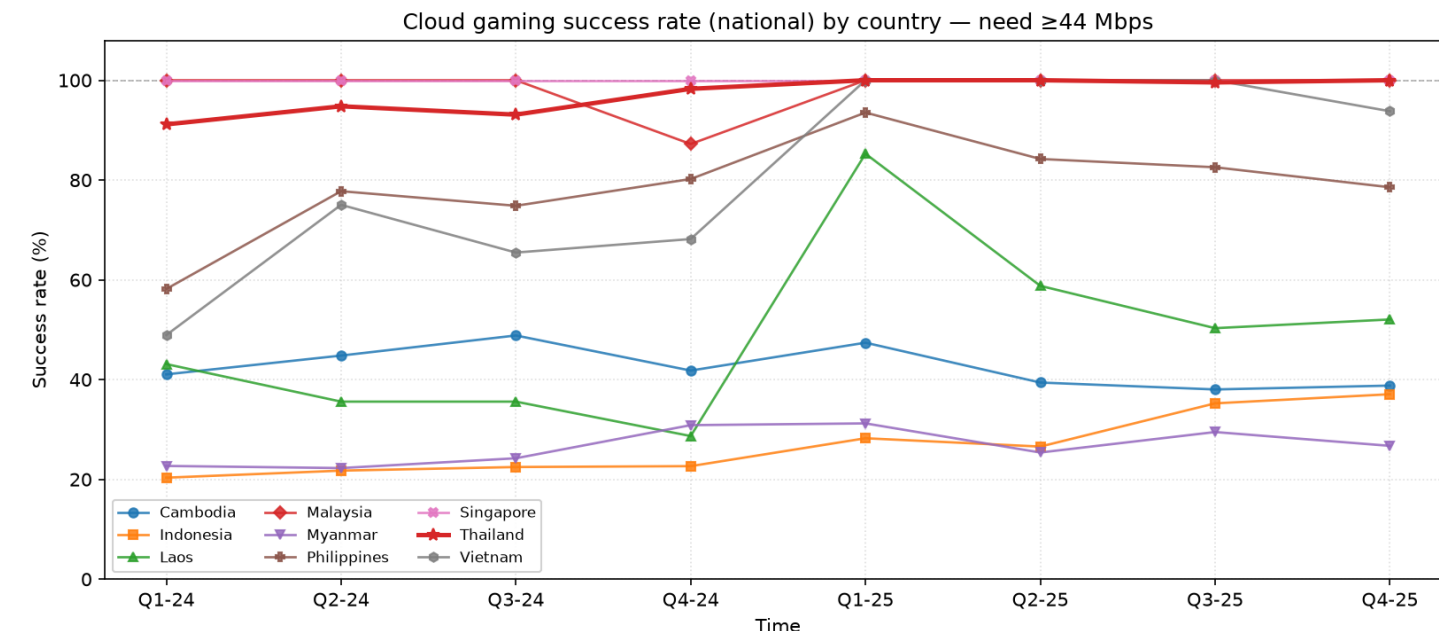
- Malaysia, Singapore, Thailand, Philippines, Vietnam: 100% every quarter
- Cambodia 67–86%; Indonesia 36 $\rightarrow$ 65%; Myanmar 40–55%
- Laos volatile: 50–100%



UHD-video success rate by country.

Cloud gaming (≥ 44 Mbps)

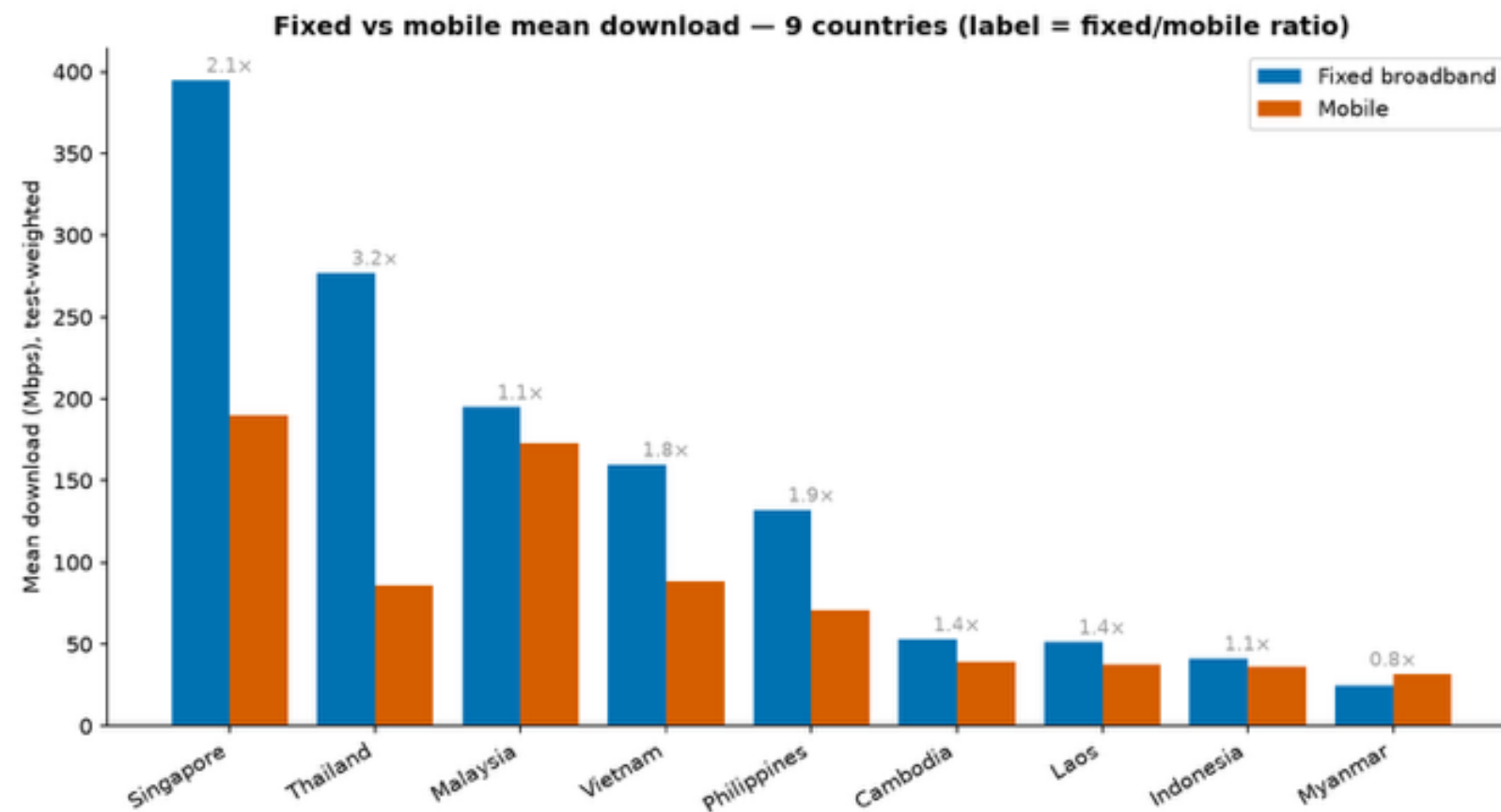
- Singapore $\sim 100\%$; Thailand and Malaysia 87–100%
- Philippines 58–93%; Vietnam $< 50\% \rightarrow \sim 94\%$
- Cambodia 38–49%; Laos $\sim 52\%$
- Indonesia $< 38\%$; Myanmar $< 31\%$



Cloud-gaming success rate by country.

Finding 5 — Mobile Is Uniform, Fixed Is Not

Fixed broadband beats mobile in most countries: Thailand shows the largest gap (**3.2 $\times$** , ~ 275 vs. ~ 85 Mbps), Singapore 2.1 $\times$, the Philippines 1.9 $\times$. Malaysia and Indonesia are near parity (1.1 $\times$). Myanmar is the only country where mobile is *faster* than fixed (0.8 $\times$) — a mobile-first market.



Fixed vs. mobile mean download by country, with ratio.

Takeaway

Raw speed rankings hide the divide that matters: not whether a country “has broadband,” but whether it survives a real application. Basic needs are met everywhere; UHD and cloud gaming expose the gap. This argues for targeted fiber deployment, regional investment and transport-layer optimization.